

One transform and one duality

a machine-checked core of music theory in Lean 4,
from pitch-class sets to tuning lattices

Carles Marín*

Independent researcher
karlesmarin@gmail.com

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Abstract

The mathematics of music theory is real mathematics, but almost all of it lives in prose and worked examples. We certify its core in Lean 4 on top of Mathlib, over the chromatic universe $\mathbb{Z}_{12}$, and we organize the library so that one object and one phenomenon do most of the work. The object is the autocorrelation, or power spectrum $|\hat{A}|^2$, of a pitch-class set: the interval vector, Babbitt’s hexachord theorem, homometry and the Z-relation read as the crystallographic phase problem, deep scales, rhythmic oddity, and the common-tone theorems are all functions of it, and maximal evenness reconnects to it in the scale theory. The phenomenon is duality: the neo-Riemannian *PLR* group is the centralizer of the regular T/I action on the 24 triads (Crans–Fiore–Satyendra, via the regular case of Wielandt’s centralizer theorem, a lemma we add over Mathlib), and the same shape recurs as a tritone-centered order-12 self-duality of the Petrushka set class 6-30 and as the val–comma lattice duality of regular temperament theory. Every theorem is sorry-free with a clean axiom audit. The underlying mathematics is classical. What we add is the formalization—to our knowledge the first comprehensive one in this area, since prior proof-assistant work covers only the T/I action—together with an autocorrelation/DFT organizing lens and two results that appear not to have been noted in the literature: the 6-30 self-duality and a unified phase taxonomy of the $|\hat{A}|^2$ invariants.

1 Introduction

What can a proof assistant certify about the mathematics of music—and how much of it is really one or two ideas seen from many sides?

Pitch-class set theory, transformational and neo-Riemannian theory, and the theory of scales and tuning have built, across half a century, a substantial body of mathematics over the cyclic group $\mathbb{Z}_{12}$, the dihedral group, $\mathbb{Z}$ -lattices, and the discrete Fourier transform: the hexachord theorem, the Z-relation, the T/I –*PLR* duality, maximal evenness, the three-gap theorem, regular temperament theory. The proofs are correct in the ordinary mathematical sense, but they are *believed* correct, not checked to the standard a kernel demands; and the literature carries a long tail of near-misses

*Prepared with the assistance of an AI system (Claude, Anthropic). The mathematics is classical; every claim, scope statement, and limitation is the author’s responsibility. The boundary of the contribution is stated plainly in Section 5, and the Lean 4 kernel—not the assistant—is what certifies the formalized parts.

and folklore stated more strongly than it holds. This paper certifies that core in Lean 4. It is the umbrella over three notes that appeared separately—on the 6-30 self-duality [29], the phase taxonomy [30], and the Tonnetz spectrum [31]—and it adds the connective tissue that makes them one library.

What machine-checking buys. Our motivation is an *honesty floor*. A machine-checked statement draws a hard line between what is proven and what is merely plausible. Where a classical argument quietly assumes a non-degeneracy, a formalization must surface it; where a textbook claim is stated for “a scale” but holds only for the diatonic, the Lean statement names the instance. We treat `UNAUDITABLE` as information, not failure: when only an instance is proven we say so, and we never inflate a first formalization into a claim of new mathematics. The contribution boundary is drawn in one place, plainly, in Section 5; everything outside it is a first formalization of a published theorem.

The model, the substrate, the trust convention. We work over $\mathbb{Z}_{12}$. A pitch-class set is a `Finset  $\mathbb{Z}_{12}$` ; a triad is a point of $\mathbb{Z}_{12} \times \text{Bool}$; a scale is a cyclically ordered tuple. We build on two pieces of Mathlib verbatim rather than rebuild them: the discrete Fourier transform `ZMod.dft` (Loeffler [1]), which gives $\hat{A}$, the power spectrum, inversion and Parseval for free, and the dihedral group `DihedralGroup 12`, which *is* the group of transpositions and inversions on the chromatic circle. Every theorem reported here is sorry-free, and we state its axiom audit: the large majority depend only on `[propext, Classical.choice, Quot.sound]`, the standard Mathlib base, with no `sorryAx`; the few finite-census results in the scale theory additionally invoke `native_decide` and so carry `Lean.ofReduceBool`, which we flag at each occurrence while keeping the surrounding analytic engines census-free.

The library at a glance. For a reader who does not work in Lean, Table 1 fixes the scale of what “machine-checked” means here. The full sources, the Sage/GAP witness scripts, and the figures are available at <https://github.com/karlesmarin/music-math>; every theorem is checkable with `#print axioms` and every script is re-runnable.

Metric	Value
Lean source files (the three pillars)	9
Theorems and lemmas	275
Total declarations (incl. defs, instances)	372
Lines of Lean	$\approx 3,800$
<code>sorry</code> / <code>admit</code>	0
New lemmas contributed to Mathlib’s gap	1 (Wielandt, regular case)
Dependency	Mathlib (Lean 4)
Axiom base	<code>propext, Classical.choice, Quot.sound</code>

Table 1: The machine-checked library behind the three pillars. Every theorem is sorry-free; the only departures from the axiom base are the finite `native_decide` censuses of §4.3, which add `Lean.ofReduceBool`.

1.1 Two recurring ideas

What makes this one paper rather than a catalogue is that two ideas organize much of it, and they recur across the three pillars (Figure 1).

Thread 1. What if the interval vector, the hexachord theorem, homometry, deep scales, rhythmic oddity, common tones, and even maximal evenness are all the same object?

The autocorrelation / DFT. A single object—the power spectrum $|\hat{A}|^2$, equivalently the autocorrelation, equivalently the crystallographers’ Patterson function—stands behind a large family of the invariants of Pillar 1 (§2). These are not analogies: each is a theorem equating the invariant to an inverse DFT of $|\hat{A}|^2$, so that homometry is *exactly* the loss of phase and the complete transform $\hat{A}$ is exactly the missing completion. The same thread resurfaces in Pillar 3: the maximally even scale is the one whose interval content extremizes a convex energy—the $|\hat{A}|^2$ -side maximizer—so the diatonic’s specialness is, once more, a statement about its autocorrelation.

Thread 2. When a group acts simply transitively, it has a dual that is its own mirror—what does that buy a chord, a symmetric chord, a tuning?

Duality: simply transitive $\Rightarrow$ self-centralizing. Lewin’s dual pair, the T/I group and the neo-Riemannian PLR group each the centralizer of the other on the 24 triads, is the Crans–Fiore–Satyendra theorem of Pillar 2 (§3); its engine is the regular case of Wielandt’s centralizer theorem. The same shape, over a different object, gives the tritone-centered order-12 self-duality of the hexachord 6-30. And it rhymes once more in Pillar 3, where regular temperament theory pairs the lattice of vals with the lattice of commas over $\mathbb{Z}$ -modules. We point out this rhyme but do not claim a meta-theorem unifying the group-action and lattice dualities; the two live in different categories, and we have no such theorem.

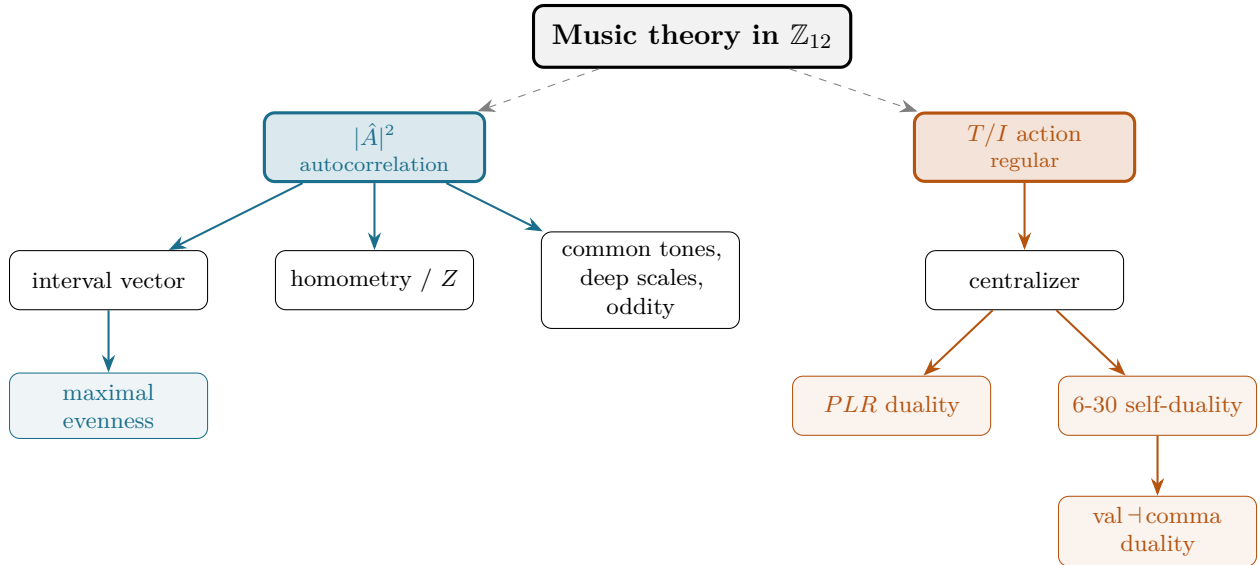


Figure 1: The thesis in one picture: the formalized core of music theory in $\mathbb{Z}_{12}$ rests on two mechanisms. *Left* — the autocorrelation $|\hat{A}|^2$ is the common root of the interval vector, homometry / the $\mathbb{Z}$ -relation, and the common-tone family (deep scales, rhythmic oddity), and through the interval vector it reaches maximal evenness (Pillars 1 and 3). *Right* — a simply transitive T/I action has a self-centralizing dual, which branches into the PLR duality of triads and the 6-30 self-duality, the latter continuing to the val $\dashv$ comma duality of temperament theory (Pillars 2 and 3). The two columns are structurally parallel; we do not claim a theorem joining them.

The arc. Pillar 1 (§2) develops the Fourier invariants and closes at the phase taxonomy. Pillar 2 (§3) develops the T/I – PLR duality and the 6-30 self-duality. Pillar 3 (§4) develops scales and tuning,

where the two threads meet again. Section 5 states plainly what is new versus first formalization; Section 6 records the open frontier—what we can state but not yet certify in the kernel.

2 Pillar 1: Fourier invariants of pitch-class sets

P₀. The interval vector, the hexachord theorem, homometry, deep scales, rhythmic oddity, the common-tone theorem — what if they are all one object, seen through different folds?

Encode a pitch-class set $A \subseteq \mathbb{Z}_{12}$ by its 0/1 indicator and take its discrete Fourier transform $\hat{A} = \mathcal{F}(\text{ind } A)$, built on Mathlib’s `ZMod.dft` (Loeffler [1]): $\hat{A}(t) = \sum_{a \in A} e^{-2\pi i at/12}$ (`Ahat`). Write $\hat{A} = |\hat{A}| e^{i\varphi}$. The claim of this pillar is that the *power spectrum* $|\hat{A}|^2$ (`powerSpec`)—equivalently the autocorrelation, equivalently the crystallographers’ Patterson function—is a single object behind a large family of the classical invariants, and that the boundary of what it can see is *exactly* the phase φ . Each statement below is a theorem in `Fourier.lean/IntervalVector.lean` (Table 2), sorry-free and axiom-clean.

2.1 The interval vector is the autocorrelation

P₁. Why is the interval vector invariant under transposition and inversion—and what is it, spectrally?

The interval vector counts, for each interval class k , the ordered pairs of A at difference class k (`IVraw`); it is invariant under transposition (`IVraw_tpose`) and inversion (`IVraw_invert`) because both only permute differences. Its spectral identity is the hub of the pillar. With the autocorrelation $\text{autocorr}(A, d) = \#\{(a, b) \in A^2 : a - b = d\}$ (`autocorr`), the Wiener–Khinchin theorem holds verbatim, $\text{autocorr}(A, d) = \mathcal{F}^{-1}(|\hat{A}|^2)(d)$ (`autocorr_eq_invDFT`), and regrading by interval class gives the keystone

$$\text{IVraw}(A, k) = \sum_{\text{ivc } d=k} \mathcal{F}^{-1}(|\hat{A}|^2)(d) \quad (\text{IVraw_eq_invDFT_power}).$$

The interval vector is therefore a function of $|\hat{A}|^2$ *alone*: it cannot see the phase. Everything in the next subsection rides on this.

2.2 The phase-blind family: hexachord, common tones, deep scales, rhythmic oddity

P₂. How much of the classical floor falls out of “it is a function of $|\hat{A}|^2$ ”?

Babbitt’s hexachord theorem. For $|A| = 6$ and every $t \neq 0$, A and its complement have opposite transforms (`Ahat_compl`), so equal power spectra (`hexachord_powerSpec_eq`) and, by the keystone, equal interval content, $\text{IVraw}(A, k) = \text{IVraw}(A^c, k)$ (`hexachord_IVraw_eq`): a hexachord and its complement are interval-equal (Babbitt; the DFT proof is Amiot [2]).

Common tones and the tritone. The pitch classes held under T_k are exactly the autocorrelation, $\#(A \cap T_k A) = \mathcal{F}^{-1}(|\hat{A}|^2)(k)$ (`commonTones_eq_invDFT`); over an interval class $1 \leq k \leq 5$ the fibre is a doubleton so $\text{IVraw}(A, k) = 2 \#(A \cap T_k A)$, while over the tritone class 6 it is a singleton and the factor 2 disappears (`IVraw_tritone`). The tritone T_6 is the same central element that organizes Pillar 2.

Deep scales and rhythmic oddity. A scale is *deep* when its six interval-class multiplicities are distinct (`IsDeep`)—by the tritone identity, exactly when every transposition retains a different number of notes (`IsDeep_iff_commonTones_nodup`); the diatonic is deep, the augmented triad is not (`diatonic_isDeep`, `augmented_not_isDeep`). A cyclic rhythm has the Rhythmic Oddity

Property when no two onsets are a tritone apart, i.e. when the autocorrelation vanishes at 6 (`rop_iff_autocorr_zero`), which forces $|A| \leq 6$ (`rop_card_le`).

2.3 Homometry: the phase the interval vector cannot see

P₃. If the interval vector is phase-blind, two sets can share it and still differ. What is the missing information?

Two sets are *homometric* when they share the autocorrelation (`Homometric`); equivalently they have the same power spectrum (`homometric_iff_powerSpec_eq`), equivalently the same diffraction intensity $|\hat{A}|^2$ at every frequency (`homometric_iff_normSq_dft_eq`). This is, verbatim, the phase problem of X-ray crystallography: the autocorrelation is the *Patterson* function (`Patterson`; Patterson 1934 [3]), $|\hat{A}|^2$ the measured intensity, homometric structures the *Z*-related chords (Forte [6]; the $Z \leftrightarrow$ Patterson correspondence is Mandereau et al. [4]). The canonical witness is the all-interval tetrachord pair (Figure 2): Homometric $\{0, 1, 4, 6\}$ $\{0, 1, 3, 7\}$ (`allIntervalTetrachords_homometric`, by `decide`) [\[b audio\]](#)—the Forte 4-Z15 / 4-Z29 pair, same magnitude, different phase, not *T/I*-related. The missing information is exactly φ .

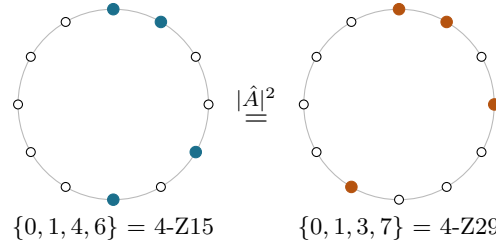


Figure 2: The *Z*-related (homometric) pair $\{0, 1, 4, 6\}$ and $\{0, 1, 3, 7\}$: identical power spectrum $|\hat{A}|^2 = [16, 2, 4, 2, 4, 2, 4, 2, 4, 2, 4, 2]$ and interval vector, yet not related by any transposition or inversion. The interval vector — a function of $|\hat{A}|^2$ alone — cannot tell them apart; only the phase of $\hat{A}$ can.

2.4 The sum vector: a first glimpse of phase

P₄. Can any invariant see some of the phase the interval vector misses?

Where the interval vector counts tones held under transposition (the difference pairing $a - b$, governed by $|\hat{A}|^2$), the *sum* / *index vector* counts tones held under inversion $I_j : x \mapsto j - x$ (the sum pairing $a + b$): $\#(A \cap I_j A) = \text{sumcorr}(A, j)$ (`commonTonesInv_eq_sumcorr`), and this is the inverse DFT of the squared *coefficient* $\hat{A}^2$, not the squared magnitude—so it retains phase (`sumcorr_eq_invDFT`). It already splits the homometric pair: about the axis $j = 0$, $\{0, 1, 4, 6\}$ [\[b audio\]](#) holds two tones and $\{0, 1, 3, 7\}$ [\[b audio\]](#) holds one (`sumVector_distinguishes_homometric`, by `decide`). The phase the interval vector cannot see, an inversive count can—partly.

2.5 The phase-taxonomy capstone

P₅. What would it take to see all of the phase?

The full transform. $\hat{A}$ is a complete invariant: $\hat{A}_A = \hat{A}_B \Rightarrow A = B$ (`Ahat_injective`), with biconditional `Ahat_eq_iff`—the proof is short and classical, since `ZMod.dft` is a `LinearEquiv` hence injective and the indicator recovers membership (`ind_injective`). Passing to a set *class*

quotients out exactly two phase operations: a transposition is a frequency-linear phase ramp on $\hat{A}$ (`tpose_iff_Ahat_ramp`) and an inversion a conjugation-plus-ramp (`inv_iff_Ahat_conj_ramp`), so B lies in the T/I orbit of A iff $\hat{A}_B$ equals $\hat{A}_A$ up to a ramp or up to conjugation-and-ramp (`setClass_iff_Ahat_mod_phase`). The pillar therefore closes as a ladder with two machine-checked boundaries certified side by side: the magnitude-only floor $|\hat{A}|^2$, where homometry collapses distinct chords (P_3), and the complete invariant $\hat{A}$, where nothing is lost (P_5). The phase is exactly the data that completes the interval vector to a total invariant; we return to what this says back to Pillar 2's self-duality in Section 3.3.

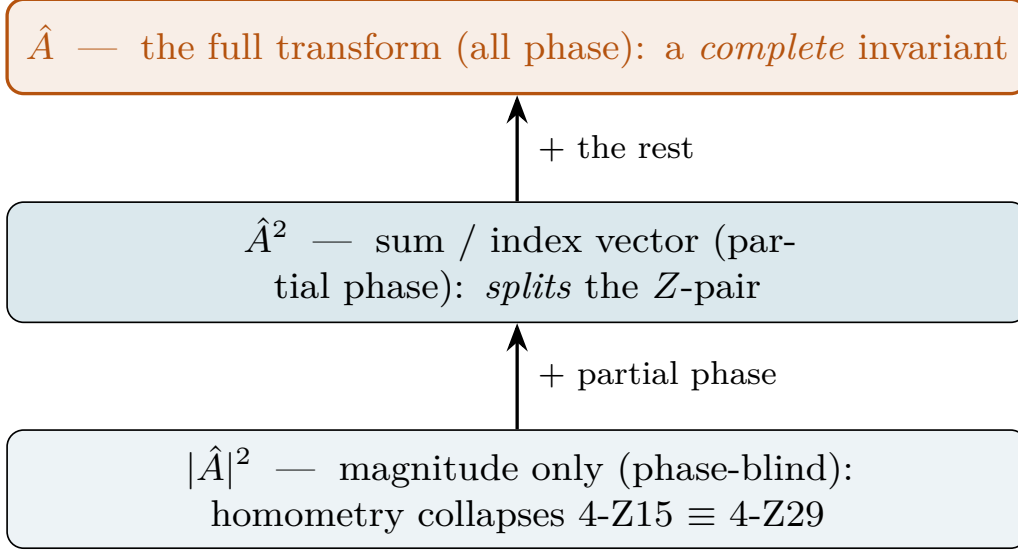


Figure 3: The taxonomy as a ladder of increasing information. The bottom rung, the power spectrum $|\hat{A}|^2$, is phase-blind, and homometry collapses the 4-Z15 / 4-Z29 pair onto it (`homometric_iff_normSq_dft_eq`); the squared coefficient $\hat{A}^2$ restores enough phase to split the pair (`sumVector_distinguishes_homometric`); the full transform $\hat{A}$ is a complete invariant (`Ahat_injective`). The two boundaries — magnitude-blind and complete — are certified side by side.

Theorem	Statement
Ahat_zero, Ahat_tpose, Ahat_compl	$\hat{A}(0) = A $; transposition = phase ramp; complement = negation off DC.
IVraw_tpose, IVraw_invert	interval vector invariant under T and I .
autocorr_eq_invDFT	Wiener–Khinchin: $\text{autocorr} = \mathcal{F}^{-1}(\hat{A} ^2)$.
IVraw_eq_invDFT_power	keystone : interval vector = $\mathcal{F}^{-1}(\hat{A} ^2)$ regraded by class.
hexachord_powerSpec_eq, hexachord_IVraw_eq	Babbitt: hexachord and complement share $ \hat{A} ^2$, hence interval content.
commonTones_eq_invDFT, IVraw_tritone	common tones = $\mathcal{F}^{-1}(\hat{A} ^2)$; tritone factor (singleton fibre).
IsDeep_iff_commonTones_nodup	deep $\iff$ distinct transposition common-tone counts.
rop_iff_autocorr_zero, rop_card_le	rhythmic oddity $\iff$ $\text{autocorr}(\cdot, 6) = 0$; forces $ A \leq 6$.
homometric_iff_normSq_dft_eq	homometry $\iff$ equal $ \hat{A} ^2$ (the discrete phase problem).
allIntervalTetrachords_homometric	$\{0, 1, 4, 6\} \sim \{0, 1, 3, 7\}$ (4-Z15 / 4-Z29), the Z -witness.
sumcorr_eq_invDFT, sumVector_distinguishes_homometric	sum vector = $\mathcal{F}^{-1}(\hat{A}^2)$ (partial phase); splits the Z -pair.
Ahat_injective, Ahat_eq_iff	the full $\hat{A}$ is a complete invariant.
setClass_iff_Ahat_mod_phase	set class = $\hat{A}$ up to a phase ramp / conjugation-and-ramp.

Table 2: Pillar 1 in `Fourier.lean` / `IntervalVector.lean`: the invariants on the ladder of phase, sorry-free with axiom audit [propxt, Classical.choice, Quot.sound] (the `decide` witnesses choice-free). Novelty is adjudicated in Section 5.

3 Pillar 2: Transformational duality

P₀. A group that acts simply transitively has a dual that mirrors it. What does that dual do for a triad—and what happens to it for a symmetric chord?

The neo-Riemannian tradition reads chords not by their notes but by the group of operations relating them, and pairs each transformation group with a dual: two subgroups of $\text{Sym}(\Omega)$ are *dual* in Lewin’s sense ([7], p. 157) when each is the centralizer of the other and both act simply transitively. This pillar machine-checks that duality for the 24 triads, then follows it into the one place it does something new—a symmetric chord, where the tritone is the center of the group.

3.1 The T/I action on triads and the PLR group

A *triad* is a pair (root, quality) $\in \mathbb{Z}_{12} \times \text{Bool}$ (major $\{r, r+4, r+7\}$ or minor $\{r, r+3, r+7\}$); there are 24. The T/I group is Mathlib’s `DihedralGroup 12`, acting on triads simply transitively—the left regular representation (`card_TIgrp` = 24, with `MulAction`, `IsPretransitive` and `FaithfulSMul` instances). The neo-Riemannian operations P, L, R are the three involutions that exchange a triad with the one sharing two common tones (`Pf, Lf, Rf`); each commutes with every T/I operation (`Pf_comm, Lf_comm, Rf_comm`), and each is a minimal voice-leading move: P and L shift one voice by a semitone (`P_size_one, L_size_one`), R by a whole tone (`R_size_two`), each holding the other two tones fixed (`P_holds_two, L_holds_two, R_holds_two`)—this is the parsimony Cohn [13] built the Tonnetz on.

3.2 PLR is the centralizer of T/I (Crans–Fiore–Satyendra)

P₁. The triad is asymmetric, so its orbit has 24 forms. Is the PLR group really the whole dual—and why?

The duality is the regular case of the classical centralizer theorem: the centralizer of a regular permutation group is its opposite regular representation (Dixon–Mortimer [15] §4.2, Cameron [16]). Its engine is a single lemma—an element of the centralizer that fixes one point is the identity (`centralizing_fixedPoint_eq_one`)—which was *absent* from Mathlib and which we supply. From

it: $\langle P, L, R \rangle$ lies in the centralizer (`PLR_le_centralizer`), the centralizer has at most 24 elements and PLR has exactly 24 (`card_PLRgrp`), and the two coincide,

$$\text{centralizer}(T/I) = \langle P, L, R \rangle \quad (\text{centralizer_eq_PLR}),$$

the Crans–Fiore–Satyendra duality [14].

3.3 The 6-30 tritone-centered self-duality

P₂. What happens to the dual when the object is a symmetric chord, whose orbit is smaller than 24?

A symmetric class has a nontrivial T/I -stabilizer, so its orbit is smaller and the action is no longer free; a reduced simply-transitive dual survives exactly when the stabilizer is *normal* in D_{12} , and among order-2 symmetries that singles out the center $\{T_0, T_6\}$ —the tritone. The unique nontrivial set class in $\mathbb{Z}_{12}$ with exactly that symmetry is Forte **6-30** = $\{0, 1, 3, 6, 7, 9\}$, the set class of Stravinsky’s Petrushka chord (Figure 4): exhaustive enumeration over the 222 nontrivial pitch-class set classes of $\mathbb{Z}_{12}$ confirms there is no other class whose T/I -stabilizer is precisely $\{T_0, T_6\}$ (a finite check, reported in Note #1 [29]). It is the union of the three central pairs $\{0, 6\}, \{1, 7\}, \{3, 9\}$, so it is fixed by the tritone T_6 (`T6_invariant`) [b audio] and has no inversional symmetry (`not_inversionally_symmetric`). Its T/I -orbit therefore collapses to 12 (`card_0`); the tritone lands in the kernel of the action on the orbit (`Ker_eq` = $\{1, r_6\}$); and the dual—the centralizer of the reduced action—has order 12 (`card_Cgrp`, built as the opposite regular representation), is transitive hence regular (`Cgrp_transitive`), and carries the dihedral fingerprint (`dihedral_fingerprint`: an order-6 generator s , an involution $t \neq 1$ with $tst = s^{-1}$ and $st \neq ts$), pinning it as dihedral of order 12. So the twelve forms of the Petrushka chord form a single navigable harmonic space [b audio], the symmetric-chord analogue of the $P/L/R$ network for triads—the tritone Stravinsky reached for by ear *is* the central symmetry T_6 , and that center is exactly what endows 6-30 with its order-12 self-duality.

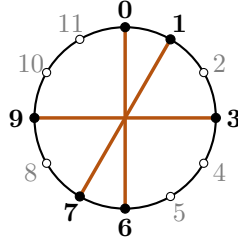


Figure 4: **6-30** = $\{0, 1, 3, 6, 7, 9\}$ on the pitch-class clock: the union of the three central pairs $\{0, 6\}, \{1, 7\}, \{3, 9\}$ (heavy tritone diameters), hence fixed by T_6 (a 180° rotation) and, with no inversional symmetry, stabilized by exactly $\{T_0, T_6\}$. That central stabilizer is what reduces the orbit to 12 and makes the dual dihedral of order 12.

3.4 The duality thread across pillars

The shape here—a simply transitive action and its self-centralizing mirror—returns in Pillar 3. There, regular temperament theory pairs the lattice of vals with the lattice of commas over $\mathbb{Z}$ -modules: the same flavour of duality, a generated structure and its annihilator. This is a structural parallel, not a theorem unifying the two (Section 5). And the self-duality of a symmetric chord is, in Fourier terms, a phase fact $|\hat{A}|$ cannot see (§2.5)— T_6 -symmetry is even-frequency support, a magnitude statement, but which symmetric chords are self-dual needs the phase, the same blindness as the Z -relation.

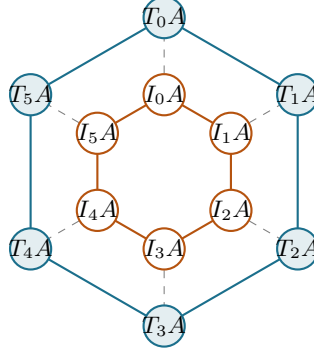


Figure 5: The order-12 dual group D_6 acting simply transitively on the 12 forms of 6-30 — its Cayley graph, the symmetric-chord analogue of the $P/L/R$ network for triads. Outer nodes are the six transposition forms $T_k A$ (with $T_k A = T_{k+6} A$ identified), inner nodes the six inversion forms $I_k A$; the solid arrows are an order-6 rotation generator (orientation reverses on the inner coset, as in any dihedral Cayley graph), the dashed edges an order-2 reflection generator. Any form of the Petrushka chord reaches any other by a unique dual move (`Cgrp_transitive`), so the twelve forms become a single navigable harmonic space.

Theorem	Statement
<i>the duality engine, NeoRiemannian.lean</i>	
<code>card_Tlgrp</code>	T/I acts on the 24 triads as the regular rep of D_{12} .
<code>Pf_comm, Lf_comm, Rf_comm</code>	P, L, R each commute with the T/I action.
<code>P_size_one, L_size_one, R_size_two</code>	minimal voice-leading sizes of P, L, R .
<code>centralizing_fixedPoint_eq_one</code>	Wielandt regular case (centralizer is semiregular) — a Mathlib gap.
<code>centralizer_eq_PLR</code>	$\text{centralizer}(T/I) = \langle P, L, R \rangle$ (CFS duality); <code>card_PLRgrp</code> = 24.
<i>the 6-30 self-duality, SixThirty.lean</i>	
<code>T6_invariant, not_inversionally_symmetric</code>	$A = \{0, 1, 3, 6, 7, 9\}$ fixed by T_6 , no inversion symmetry.
<code>card_0, Ker_eq</code>	orbit 12; kernel of the action = $\{T_0, T_6\} = Z(D_{12})$.
<code>card_Cgrp, Cgrp_transitive</code>	dual (centralizer on the orbit) order 12, regular.
<code>dihedral_fingerprint</code>	order 12; $s^6=1$, $t^2=1 \neq t$, $tst=s^{-1}$, $st \neq ts$ (dihedral).

Table 3: Pillar 2 in `NeoRiemannian.lean` / `SixThirty.lean`, sorry-free with axiom audit [propext, Classical.choice, Quot.sound]. The 6-30 dual is the one place the paper goes beyond formalization; its scope and the relation to the Fiore lineage are stated in Section 5.

4 Pillar 3: scales and tuning

P_0 . *Why is the diatonic scale special—and is its specialness one fact, or several that happen to coincide?*

Four regularity conditions on a scale—being generated by a single interval, being well-formed, having Myhill’s property, being maximally even—pick out the diatonic, and the last reconnects to the autocorrelation of Pillar 1. This pillar machine-checks each, names the instance where only an instance is proven, and ends at the tuning lattices where the duality thread of Pillar 2 returns.

4.1 Generated and well-formed scales

Stacking k perfect fifths from a starting pitch and reducing mod 12 (`stack`) generates the diatonic at $k = 7$ (`diatonic_generated_by_fifth`) and the pentatonic at $k = 5$ (`pentatonic_generated_by_fifth`). A scale is *well-formed* when it has exactly two step sizes (`IsWellFormed`). The diatonic is well-formed, with steps $\{1, 2\}$ semitones (`diatonic_isWellFormed`,

diatonic_stepSizes), and so is the pentatonic, with steps $\{2, 3\}$ (`pentatonic_isWellFormed`); stacking too far breaks it — six fifths give three step sizes (`sixFifths_not_wellFormed`). That a generated scale has at most *three* step sizes is the three-gap (Steinhaus) theorem, which we verify in $\mathbb{Z}_{12}$ (`threeGap_chromatic`; Sós [26]). Carey–Clampitt’s well-formedness theory [22] is the frame.

4.2 Myhill’s property

P₁. Well-formedness counts step sizes. Myhill’s property counts something finer—and it separates the diatonic from its even cousin.

A scale has Myhill’s property when every nonzero generic interval (second, third, ...) comes in exactly two specific sizes (`HasMyhill`). The diatonic does (`diatonic_hasMyhill`); the whole-tone scale does *not* (`wholeTone_not_hasMyhill`), its single whole-step collapsing the variety. By Carey–Clampitt, Myhill is equivalent to non-degenerate well-formedness; we formalize the diatonic instance and the whole-tone counterexample, and note the general equivalence (which needs Christoffel/three-gap word machinery, thin in Mathlib) as not yet formalized.

4.3 Maximal evenness

P₂. “Spread k notes as evenly as possible”: which energy makes that precise, and does it actually single out the diatonic?

Maximal evenness (Clough–Douthett [23]) is the deepest of the four conditions, and we formalize it in three layers, explicit about which energy does the work [b audio]. (i) *The predicate.* A scale is maximally even when each generic interval takes at most two specific sizes and those are consecutive integers (`IsMaxEven`); the diatonic and the whole-tone scale both qualify (`diatonic_isMaxEven`, `wholeTone_isMaxEven`), so maximal evenness strictly contains Myhill (the whole-tone scale is maximally even but, by §4.2, not Myhill). (ii) *The step-gap energy.* For a strictly convex potential V , the energy $\sum_i V(\text{gap}_i)$ over the circular step gaps is minimized exactly at the *balanced* configurations, $\max - \min \leq 1$ (`isMinimizer_iff_balanced`), via a strict-convexity swap lemma and a well-founded descent. This characterizes the balanced step multiset—a *necessary* condition for maximal evenness, but not sufficient, and we say so in Section 5. (iii) *The all-pairs energy.* The energy that uniquely selects the diatonic is the Douthett–Krantz all-pairs Hamiltonian $E(V, A) = \sum_{a < b} V(\text{cdist}(a, b))$ over all pairwise distances. For every decreasing-convex V the diatonic minimizes it, and for every strictly such V it is the *unique* minimizer up to transposition and inversion (`diatonic_ground_state`, `diatonic_unique`)—one statement holding simultaneously for all admissible potentials, the V -independence. The engine is a double summation-by-parts identity (`abel_bridge`): two k -subsets have equal interval-vector totals $\binom{k}{2}$, so their difference sums to zero and the energy gap collapses to a sum of double-cumulative sums against the convexity of V . The same theorem holds for the other maximally even cardinalities in $\mathbb{Z}_{12}$ — pentatonic ($k = 5$), whole-tone ($k = 6$), octatonic ($k = 8$), joining the diatonic ($k = 7$) — (`pentatonic_unique`, `wholetone_unique`, `octatonic_unique`), via a cardinality-independent core. (Equivalently the diatonic uniquely maximizes $|\hat{A}(5)|$, Quinn [5]—Pillar 1’s autocorrelation thread, returning.)

4.4 Regular temperament theory: vals and comma lattices

P₃. Twelve fifths nearly close the octave; the gap is the Pythagorean comma. What is the algebra of “nearly”?

Model a pitch by its exponents in the prime generators: a val is a $\mathbb{Z}$ -linear map sending generators to step counts (`v12`, `v5`, `v7`). Twelve fifths minus seven octaves is the Pythagorean

comma pc (`closure_defect_eq_pc`); 12-tone equal temperament is exactly the val that annihilates it, and the comma generates the whole rank-1 kernel, $\ker v_{12} = \mathbb{Z} \cdot \text{pc}(\text{ker_v12_eq_span_pc})$, with only the 12-val killing it (`only_twelve_tempers_pc`). In the 5-limit the syntonic comma enters: 12-ET tempers it (`v12_5_tempers_syntonic`), four fifths equal a major third up to that comma (`meantone_defect_eq_syntonic`)—so 12-ET is a meantone—and the comma kernel is rank 2, spanned by the syntonic and Pythagorean commas (`ker_v12_5_eq_span`). Reducing the fifth mod 12 closes the circle of fifths onto all twelve pitch classes (`circle_of_fifths_complete`, Figure 6), and four fifths reduce to the major third (`four_fifths_eq_major_third`). The val–comma pairing—a generated lattice and its annihilator—is the $\mathbb{Z}$ -module echo of Pillar 2’s centralizer duality, the rhyme of §3.4.

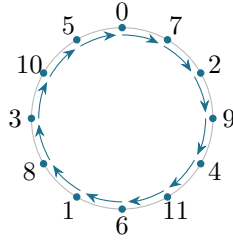


Figure 6: The circle of fifths: stepping by the reduced fifth (7 semitones) visits all twelve pitch classes (`circle_of_fifths_complete`); the node labels are the pitch classes reached in fifth-order. Closure is the rank-1 comma kernel of `ker_v12_eq_span_pc`.

Theorem	Statement
<code>diatonic_generated_by_fifth</code> , <code>pentatonic_generated_by_fifth</code>	diatonic / pentatonic generated by stacked fifths.
<code>diatonic_isWellFormed</code> , <code>sixFifths_not_wellFormed</code>	well-formed = two step sizes; the boundary.
<code>threeGap_chromatic</code>	three-gap / Steinhaus theorem in $\mathbb{Z}_{12}$.
<code>diatonic_hasMyhill</code> , <code>wholeTone_not_hasMyhill</code>	Myhill’s property: diatonic yes, whole-tone no.
<code>diatonic_isMaxEven</code> , <code>wholeTone_isMaxEven</code>	maximal evenness (predicate); $\text{ME} \supsetneq \text{MP}$.
<code>isMinimizer_iff_balanced</code>	step-gap energy minimizer $\iff$ balanced multiset (necessary, not sufficient).
<code>diatonic_ground_state</code> , <code>diatonic_unique</code>	all-pairs energy: diatonic the unique minimizer up to T/I , $\forall$ admissible V .
<code>pentatonic_unique</code> , <code>wholetone_unique</code> , <code>octatonic_unique</code>	same, at $k \in \{5, 6, 8\}$, via <code>abel_bridge</code> .
<code>closure_defect_eq_pc</code> , <code>ker_v12_eq_span_pc</code>	Pythagorean comma = closure defect; rank-1 kernel.
<code>meantone_defect_eq_syntonic</code> , <code>ker_v12_5_eq_span</code>	syntonic comma / meantone; rank-2 5-limit kernel.
<code>circle_of_fifths_complete</code> , <code>four_fifths_eq_major_third</code>	twelve fifths generate $\mathbb{Z}_{12}$; 4 fifths = major third.

Table 4: Pillar 3 in `DiatonicScale.lean`, `MaximalEvenness.lean`, `AllPairsEvenness.lean`, `Temperament.lean`, `CircleOfFifths.lean`, sorry-free; axiom audit [propext, Classical.choice, Quot.sound], except the finite all-pairs censuses, which additionally carry `Lean.ofReduceBool` (`native_decide`), flagged in Section 5.

5 Novelty, scope, and what this work does not claim

What here is new mathematics, what is a first formalization of known mathematics, and what is merely organization? We answer in one place, ruthlessly, so the framing cannot drift.

Contribution, stated plainly. This is a formalization paper. With the two exceptions below, every result is a machine-checked proof of a published theorem — to our knowledge a first formalization — cited to its source in its pillar and collected in Table 5. The library is sorry-free; the axiom audit is [propext, Classical.choice, Quot.sound] throughout, save the finite maximal-evenness censuses, which additionally carry Lean.ofReduceBool from `native_decide`—flagged here as in the body, with the analytic engine behind them (a double summation-by-parts identity) kept axiom-clean.

What is genuinely ours (1): the 6-30 self-duality. The one place where we believe we go beyond formalization is the tritone-centered order-12 self-duality of the Petrushka set class $6-30 = \{0, 1, 3, 6, 7, 9\}$ (§3.3). It is not inversionally symmetric yet is fixed by the tritone T_6 ; its T/I -orbit therefore collapses to 12, the tritone lands in the kernel of the action, and the centralizer of the reduced action is dihedral of order 12. We are unaware of a previous explicit characterization of this dual, or of a machine-checked contextual dual of a symmetric chord. We state it *measuredly*, as in Note #1 [29]: it is *complementary* to the Fiore lineage rather than something they missed. Fiore–Satyendra [19] and Fiore–Noll [20, 21] each sidestep the reduced-orbit tritone-central regime; [20] even tabulates the same set class as a point and studies its order-2 cover, never forming the order-12 dual of the hexachord-as-a-point. Berry–Fiore [17] prove the same centralizer duality for hexatonic cycles—a different object (augmented stabilizer, dual S_3 of order 6). The one residual collision risk, Peck’s generalized commuting groups [18], is hedged, not dismissed; the published octatonic commuting groups in this lineage are order 8, not 12. What we claim is the explicit characterization and its machine-checked dual, not priority over the general theory.

What is genuinely ours (2): the unified phase taxonomy. Every ingredient of the phase taxonomy is classical (Patterson, Lewin, Babbitt, Amiot). What we believe is unstated *as a single statement* is the unification: pinning each invariant to its exact slice of $\hat{A}$ and certifying both boundaries of the ladder—the magnitude-blind collapse (homometry) and the complete-invariant capstone—side by side, with one axiom audit (§2.5). This is organization, not new mathematics; we claim priority over none of the underlying results.

A smaller framing point. In the scale theory we make explicit that the convex-energy characterization of maximal evenness splits into a degenerate step-gap form and a diatonic-selecting all-pairs form (§4.3); this is a clarification of which object does the work, not a theorem.

Prior proof-assistant work. The only prior Lean music formalization we are aware of (Prism-river [28]) addresses the T/I action; the three layers here—the DFT / interval-vector invariants, the neo-Riemannian duality, and scale and tuning theory—are, to our knowledge, first formalizations. One reusable group-theory lemma, the regular case of Wielandt’s centralizer theorem (`centralizing_fixedPoint_eq_one`), was absent from Mathlib and is supplied here.

Result	Classical source	Status
<i>Pillar 1 — Fourier invariants</i>		
Interval vector = $\mathcal{F}^{-1} \hat{A} ^2$; common tones	Amiot [2]; Lewin [7], Rahn [10]	first-formaliz.
Babbitt’s hexachord theorem	Babbitt; Lewin [9]	first-formaliz.
Deep scales; rhythmic oddity	Gamer/Browne [11]; Chemillier–Truchet [12]	first-formaliz.
Homometry = phase problem	Patterson [3]; Forte [6]	first-formaliz.
Unified phase taxonomy (one statement)	— (this work)	ours: organization
<i>Pillar 2 — Transformational duality</i>		
T/I regular rep; PLR voice-leading	Lewin [7]; Cohn [13]	first-formaliz.
Wielandt regular-case lemma	Dixon–Mortimer [15]	first-formaliz. [†]
CFS duality: $PLR = \text{centralizer}(T/I)$	Crans–Fiore–Satyendra [14]	first-formaliz.
6-30 tritone-centered self-duality	— (this work)	ours: new result
<i>Pillar 3 — Scales and tuning</i>		
Myhill; well-formed; three-gap	Carey–Clampitt [22]; three-gap	first-formaliz.
Maximal evenness (predicate; energy)	Clough–Douthett [23]; Douthett–Krantz [24]	first-formaliz.
step-gap vs all-pairs distinction	— (this work)	ours: organization
Comma lattices (Pythagorean, syntonic)	regular temperament theory	first-formaliz.

Table 5: The honesty ledger. Every row is a first formalization of classical mathematics, cited to its source, except the two **ours** rows (one new result, one organization) and the smaller framing point. [†] The Wielandt regular-case lemma was absent from Mathlib; supplying it is a small reusable formalization contribution beyond the music content.

6 The open frontier

What can we state but not yet certify in the kernel—and where does each thread point next?

In the spirit of the wider program, we treat the boundary of what is formalized as an output, not a gap to be hidden. Four directions stand open.

- **The general- n self-dual family.** The reduced-duality characterization behind 6-30 is universe-independent; 6-30 is the minimal nontrivial instance of a family with dual $\cong D_{n/2}$ at universe n (the count is OEIS A032239). A uniform Lean statement, and the named isomorphism $D_{12}/Z \cong D_6$ (currently a Mathlib gap, witnessed in GAP), are open.
- **Maximal evenness in general (k, n) .** The all-pairs V -independent uniqueness is certified over $\mathbb{Z}_{12}$ for $k \in \{5, 6, 7, 8\}$ by finite census; the analytic engine (the double summation-by-parts identity) is general, but the general- (k, n) theorem needs the abstract majorization argument, and replacing the censuses by a kernel proof would retire the lone `Lean.ofReduceBool` dependence.
- **Phase and self-duality.** T_6 -symmetry is exactly even-frequency support of $\hat{A}$, a magnitude fact; but self-duality needs phase information $|\hat{A}|$ cannot see—the same blindness as the Z -relation. A phase-space account of self-duality, and a Lean theorem for set-class completeness (currently computational), are open.
- **From analysis to composition.** The library is an analysis engine; the maximally even generator, the well-formed phase diagram, and the dual-group navigation of a symmetric chord are constructive and point toward a synthesis program downstream.

Data and code availability

The Lean sources, the Sage/GAP witness scripts, and the figures accompany this paper; each Lean theorem is checkable by `#print axioms` and each script is re-runnable. The three constituent notes are archived on Zenodo ([29, 30, 31]).

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